



University of Asia Pacific

Department of Computer Science and Engineering

CSE 316: Microprocessors and Microcontrollers Lab

LAB REPORT

Experiment Number:06

Experiment Title: Wireless Theft Detection using Door Sensor and Buzzer

Submitted by:

Name : Sunandan Das

Student ID : 23101218

Section : D2

Submitted to:

Zaima Sartaj Taheri

Lecturer,

Department of Computer Science and Engineering

Date of Submission: 17/04/2026

1. Experiment Name : Wireless Theft Detection using Door Sensor and Buzzer

2. Objective:

The objectives of this project are to design and implement a wireless theft detection system using a PIR motion sensor, buzzer, and LED interfaced with an Arduino Uno. The project aims to detect unauthorized motion or entry near a door and trigger an audible and visual alarm to alert users. Through this experiment, the practical application of PIR sensors in motion-based security systems is explored, demonstrating how sensor inputs can be processed by a microcontroller to control output devices. Furthermore, the experiment provides hands-on experience in integrating sensors and actuators with Arduino, enhancing practical knowledge in embedded systems and home security automation.

3. Hardware & Software Requirements:

Hardware:

- 1 × Arduino Uno R3
- 1 × PIR Motion Sensor
- 1 × Buzzer
- 1 × LED
- 1 × Resistor (220Ω)

Software:

- Arduino IDE
- Tinkercad

4. Circuit Diagram:

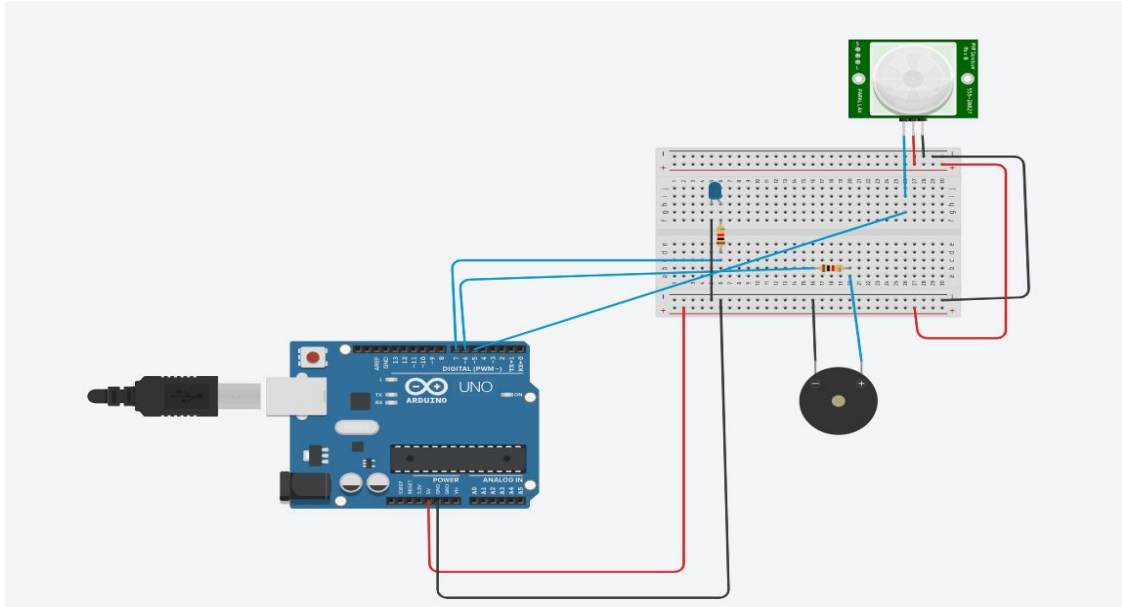


Figure 1: Wireless Theft Detection System Circuit Design

5. Assembly Program:

```
// Wireless Theft Detection using Door Sensor and Buzzer
```

```
const int pirPin = 5;
```

```
const int buzzerPin = 6;
```

```
const int ledPin = 7;
```

```
int pirState = LOW;
```

```
int val = 0;
```

```
void setup() {
```

```
    pinMode(pirPin, INPUT);
```

```
    pinMode(buzzerPin, OUTPUT);
```

```
    pinMode(ledPin, OUTPUT);
```

```
    Serial.begin(9600);
```

```
}
```

```
void loop() {
```

```
    val = digitalRead(pirPin);
```

```
    if (val == HIGH) {
```

```
        if (pirState == LOW) {
```

```
            Serial.println("Motion detected!");
```

```
            Serial.println(val);
```

```
            digitalWrite(buzzerPin, HIGH);
```

```
            digitalWrite(ledPin, HIGH);
```

```
            pirState = HIGH;
```

```
}  
  
} else {  
  
    if (pirState == HIGH) {  
  
        Serial.println("Motion ended");  
  
        Serial.println(val);  
  
        digitalWrite(buzzerPin, LOW);  
  
        digitalWrite(ledPin, LOW);  
  
        pirState = LOW;  
  
    }  
  
}  
  
  
delay(200);  
}
```

6. Observations:

Upon simulating the Wireless Theft Detection System in Tinkercad, the following observations were noted:

a) PIR Motion Detection:

- The PIR sensor effectively detected motion within its designated range and generated a HIGH signal.
- In the absence of motion, the signal remained LOW, indicating a secure state. - The simulated detection range in Tinkercad was consistent and reliable for virtual movement.

b) LED and Buzzer Response:

- When motion was detected, both the LED and buzzer were activated simultaneously, simulating an alarm system.
- When no motion was present, the LED and buzzer remained OFF.
- The system responded immediately to changes in the PIR sensor output.

c) Circuit Behavior:

- The Arduino continuously monitored the PIR sensor output and updated the alarm status accordingly.
- The Serial Monitor provided messages such as “Motion detected!” and “Motion ended,” in real time, facilitating easy debugging and verification of system functionality.

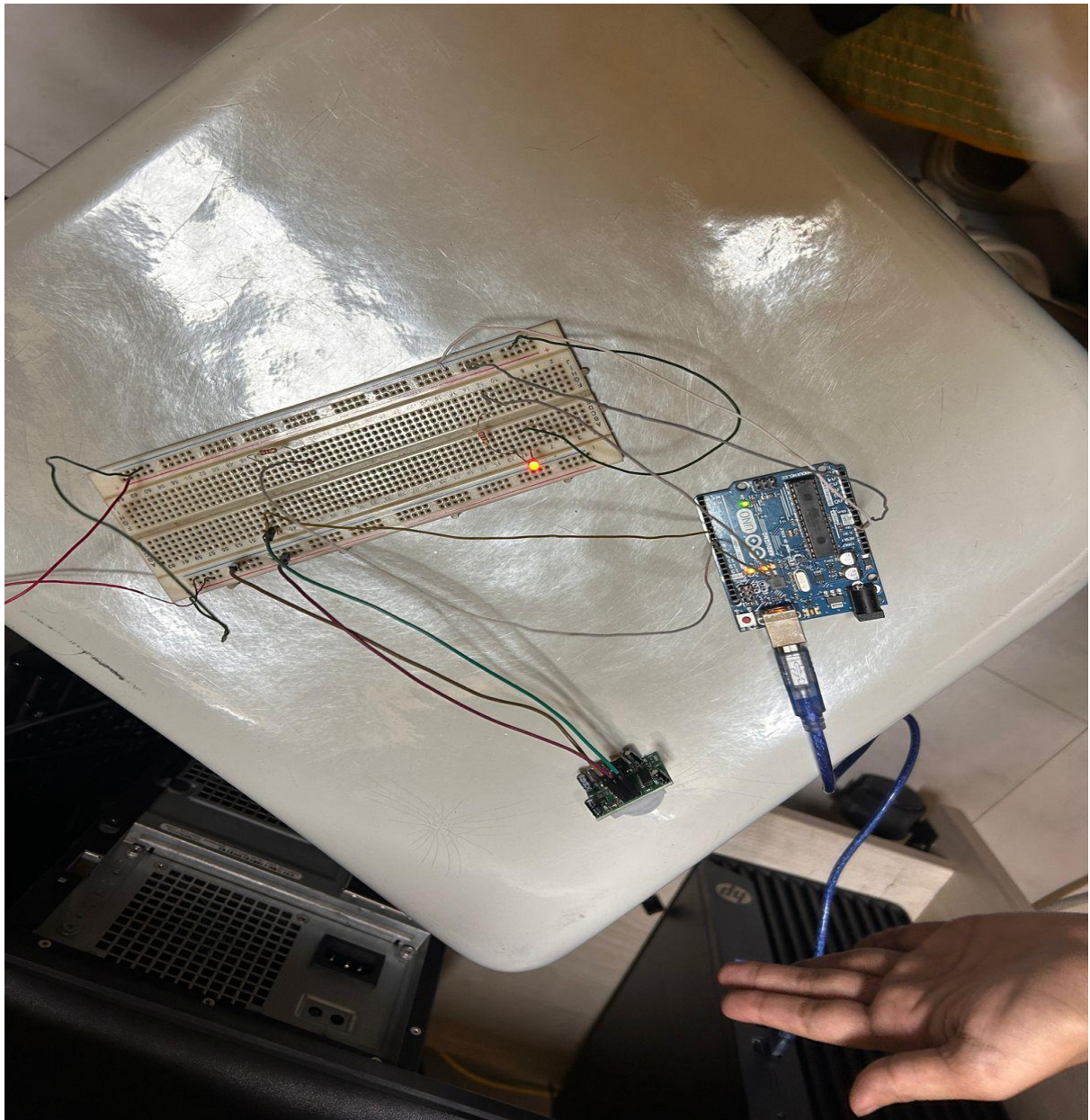


Figure 2: Output On Serial Monitor Upon Motion Detection

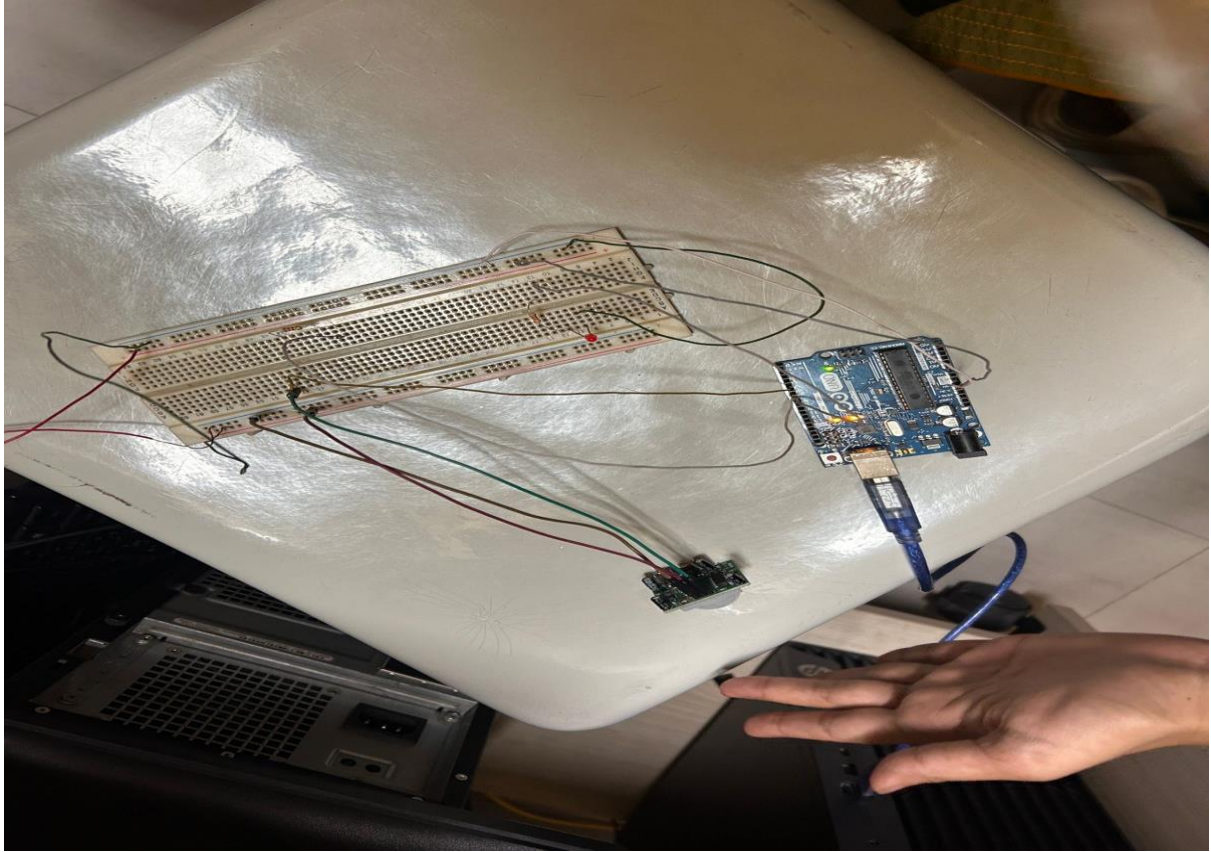


Figure 3: Output On Serial Monitor Upon No Motion Detection

- The system successfully demonstrated basic intrusion detection logic suitable for home security applications.

d) System Sensitivity and Range:

- The PIR sensor was tested in the simulation for various distances and angles of movement.
- Motion was reliably detected within a virtual sensing range of approximately 3–5 meters.
- The system responded promptly to any movement in front of the sensor, indicating good sensitivity.
- As this was a virtual simulation, actual range and response time may vary when implemented in hardware.

7. Result:

A simulated Wireless Theft Detection System effectively identified motion through the use of a PIR sensor. The Arduino promptly processed the sensor signals and activated both the LED and buzzer, providing immediate visual and audible alerts. This reliable and instantaneous response confirmed the correctness of the motion-triggered alarm logic. Ultimately, the system effectively demonstrated how a microcontroller-based motion detection and alarm mechanism can be implemented using simple sensors and actuators, showcasing practical applications for home security automation.

8. Conclusion:

This experiment effectively presented the development and simulation of a motion-triggered theft detection system using Arduino and PIR sensors. The setup successfully employed the PIR sensor to sense infrared radiation caused by human movement and converted its digital output to trigger both the LED and buzzer, ensuring instant visual and audible warnings. The system demonstrated the practical application of motion sensing in security and automation, the integration of multiple output components with a microcontroller, and the implementation of continuous monitoring and decision-making logic in Arduino. Although the Tinkercad simulation could not perfectly represent real-world conditions such as precise detection range, environmental disturbances, or variations in buzzer intensity and response delay, it clearly illustrated the core working principles and overall functionality of an Arduino-based theft detection system.